LPBF optimization is not a 2-parameter problem

A *Power/scan-speed studies explore 2–4 variables, while practical LPBF control spans 36+ dimensions — and grows with each added material*

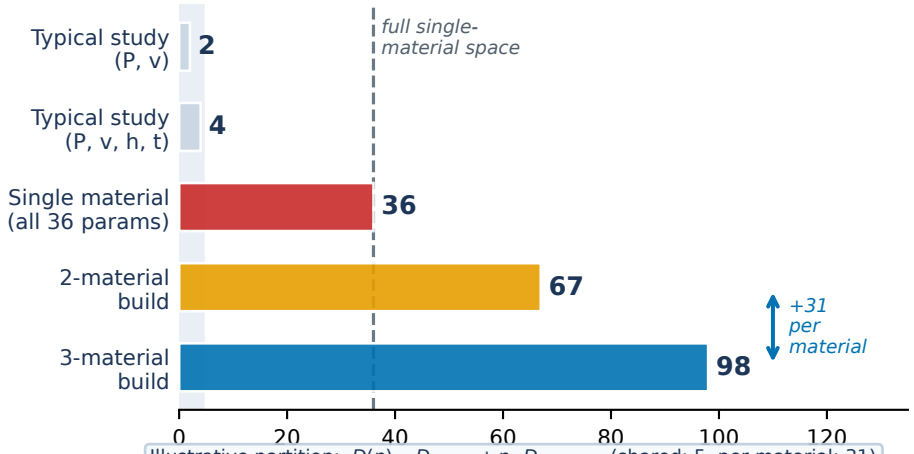
36 controllable variables across beam, scan, powder, environment, and machine domains

Category (A dots / C nodes): Beam · Scan Strategy · Layer · Powder · Atmosphere · Thermal Mgmt · Machine-Specific

Beam *8 params*	**Laser power**	**Scan speed**	Spot size	Beam profile	Wave-length	CW/ Pulsed	Pulse params	Beam oscill.
Scan *8 params*	**Hatch spacing**	Scan pattern	Rotation angle	Vector length	Contour params	Contour passes	Sky-writing	**Scan vs. gas flow**
Layer *4 params*	**Layer thick.**	Build orient.	Support strategy	**Plate location**				
Powder *6 params*	PSD	Morpho-logy	Flowa-bility	Packing density	**Moisture content**	**Recycling state**		
Atmos. *4 params*	Gas type	**Gas flow rate**	O_2 content	Chamber pressure				
Thermal *3 params*	Baseplate preheat	**Inter-lyr dwell**	Build height					
Machine *3 params*	Recoater type	Recoater speed	Dosing factor					

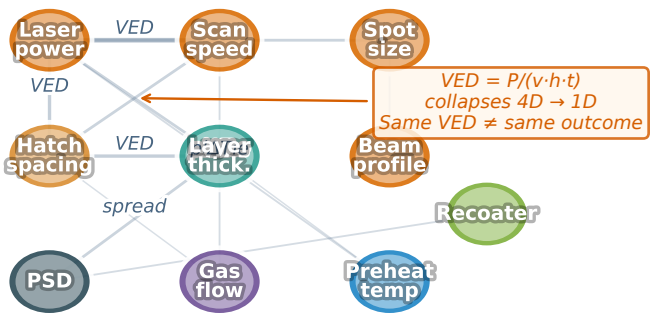
36 parameters

B Published studies explore 2–4 dimensions; real search spaces are an order of magnitude larger



Illustrative partition: $D(n) = D_{shared} + n \cdot D_{material}$ (shared: 5, per-material: 31)

C Cross-category couplings make 1D surrogates like VED insufficient



Schematic coupling map; edge weight is qualitative

36–98D search spaces with strong coupling are intractable for conventional DoE → autonomous Bayesian optimization is well-suited

Optimization prevalence (Panel A accent color)

Actively optimized (4) · Selectively optimized (9) · Typically fixed (17) · Often neglected (6)

Literature synthesis (38 studies) | Oliveira et al. 2020, Buhairi et al. 2023, Ferro et al. 2020